

Composition of transformations vs. exponential map

This example explores the difference between MLS eq. (2.24) p. 37 and MLS eq. (2.37) p. 42.

The example uses material taken from question 2 if Exam A of 2025.

The origin of the body frame B of a rigid body has coordinates $[1.8 \ -0.2 \ 0.05]^T$ in the inertial frame A . The directions of the axes of frame B with respect to the inertial frame is defined by the rotation matrix R_{ab} . The components of R_{ab} is given to you in double precision in the Excel spreadsheet file “q2dat.xlsx”, in the “Exam_A_files.zip” file on Moodle, under the “Exam” topic, or as indicated below:

9.1168461167710357e-01	3.2205442693477448e-01	-2.5517075640222342e-01
-2.2792115291927589e-01	9.1309336395051188e-01	3.3810125223294596e-01
3.4188172937891381e-01	-2.5008289585905735e-01	9.0585618522789324e-01

If you choose to read the data from the file into MATLAB, you can use the command

```
Rab = readmatrix('q2dat.xlsx')
```

- (a) The body is now displaced so that the origin of the body frame has coordinates $[0 \ -1.2 \ -0.05]^T$ in the inertial frame, while its x and y axes point toward the negative x and negative y directions of the inertial frame, respectively. To avoid confusion, we refer to the body frame after this displacement as frame C . Calculate the rigid transformation matrix $g_{bc} \in SE(3)$ associated with the motion described above.

```
clearvars, clc
Rab = readmatrix('q2dat.xlsx')
```

```
Rab = 3x3
    0.9117    0.3221   -0.2552
   -0.2279    0.9131    0.3381
    0.3419   -0.2501    0.9059
```

```
gab = [Rab,[1.8;-0.2;0.05];zeros(1,3),1]
```

```
gab = 4x4
    0.9117    0.3221   -0.2552    1.8000
   -0.2279    0.9131    0.3381   -0.2000
    0.3419   -0.2501    0.9059    0.0500
         0         0         0    1.0000
```

```
x2 = [-1;0;0];
y2 = [0;-1;0];
z2 = cross(x2,y2)
```

```
z2 = 3x1
    0
    0
    1
```

```
Rac = [x2,y2,z2]
```

```
Rac = 3x3
    -1     0     0
     0    -1     0
     0     0     1
```

```
gac = [Rac,[0;-1.2;-0.05];zeros(1,3),1]
```

```
gac = 4x4
    -1.0000     0     0     0
         0    -1.0000     0    -1.2000
         0     0     1.0000    -0.0500
         0     0     0     1.0000
```

MLS eq. (2.24) p. 37 (composition of two transformations):

$$g_{ac} = g_{ab}g_{bc}$$

Thus,

$$g_{bc} = g_{ab}^{-1}g_{ac}$$

```
gbc = gab\gac
```

```
gbc = 4x4
    -0.9117     0.2279     0.3419    -1.4473
    -0.3221    -0.9131    -0.2501    -1.4678
     0.2552    -0.3381     0.9059     0.0306
         0         0         0     1.0000
```

If I wish to express the rigid body motion from B to C in terms of a exponential map for a twist, I need to use MLS eq. (2.37) p. 42:

$$g_{ac} = e^{\hat{\xi}\theta}g_{ab} = g(\theta)g_{ab}$$

where the notation is introduced here that $g(\theta) = e^{\hat{\xi}\theta}$, similarly to the notation used in MLS example 2.2 p. 43. In the sense of eq. (2.37), one can think of g_{ab} as the initial configuration of the rigid body. Therefore,

$$g(\theta) = g_{ac}g_{ab}^{-1}.$$

```
gtheta = gac/gab
```

```
gtheta = 4x4
    -0.9117     0.2279    -0.3419     1.7037
    -0.3221    -0.9131     0.2501    -0.8154
    -0.2552     0.3381     0.9059     0.4316
         0         0         0     1.0000
```

```
dum1 = rotm2axang(gtheta(1:3,1:3)) % MATLAB function alternative to omeg_thet
```

```
dum1 = 1x4
    0.1561    -0.1538    -0.9757     2.8559
```

```
omega = dum1(1:3)'
```

```
omega = 3x1
    0.1561
   -0.1538
   -0.9757
```

```
theta = dum1(4)
```

```
theta =  
2.8559
```

The next cell tests whether the calculated variables ω ("omega") and θ ("theta") are correct.

```
gtheta(1:3,1:3);  
h_omega = hatm_num(omega);  
shouldbezero4times4 = gtheta(1:3,1:3) - rodrigues(h_omega,theta)
```

```
shouldbezero4times4 = 3×3  
10-15 ×  
    0.3331    0.4996    0.3886  
   -0.0555    0.5551   -0.6661  
    0.4441   -0.6106    0.2220
```

The next cell calculates v , using MLS eqn. (2.38) p. 43.

```
A = (eye(3) - gtheta(1:3,1:3))*h_omega + omega*omega'*theta
```

```
A = 3×3  
    0.3446    1.8500   -0.6936  
   -1.9736    0.3428    0.0804  
   -0.0907    0.6923    2.7322
```

```
v = A\gtheta(1:3,4)
```

```
v = 3×1  
    0.5522  
    0.8075  
   -0.0283
```

```
shoudlbezero3times1 = gtheta(1:3,4) - A*v
```

```
shoudlbezero3times1 = 3×1  
10-16 ×  
     0  
     0  
   -0.5551
```

The next cell calculates the twist as a 6×1 vector.

```
format long  
xi = [v;omega]
```

```
xi = 6×1  
    0.552235188427939  
    0.807464983844195  
   -0.028279569457120  
    0.156148523245089  
   -0.153829166292768  
   -0.975681416388592
```

```
format default
```

The next cell test whether the twist (v, ω) , or (v, ω) in MATLAB variables, does actually produce $g(\theta)$.

```
dummat = gtwist_num(xi,theta)
```

```
dummat = 4×4  
   -0.9117    0.2279   -0.3419    1.7037  
   -0.3221   -0.9131    0.2501   -0.8154  
   -0.2552    0.3381    0.9059    0.4316
```

```
0      0      0      1.0000
```

```
gtheta
```

```
gtheta = 4x4
-0.9117    0.2279   -0.3419    1.7037
-0.3221   -0.9131    0.2501   -0.8154
-0.2552    0.3381    0.9059    0.4316
0          0          0          1.0000
```

```
mustbezero4times4 = dummat - gtheta
```

```
mustbezero4times4 = 4x4
10-15 x
-0.3331   -0.4996   -0.3886   -0.2220
 0.0555   -0.5551    0.6661   -0.2220
-0.4441    0.6106   -0.2220    0.8327
0          0          0          0
```

Comparison between g_{bc} and $g(\theta)$

The interpretation of g_{ab} is as follows:

- its upper left submatrix $R_{ab} \in \mathbb{R}^{3 \times 3}$ is the rotation matrix in components in frame A that generates frame B .
- its 4th column p_{ab} is the homogeneous coordinates of the frame B origin expressed in frame A .

Likewise, the interpretation of g_{ac} is as follows:

- its upper left submatrix $R_{ac} \in \mathbb{R}^{3 \times 3}$ is the rotation matrix in components in frame A that generates frame C .
- its 4th column p_{ac} is the homogeneous coordinates of the frame C origin expressed in frame A .

What is perhaps less obvious is that the interpretation of g_{bc} is as follows:

- its upper left submatrix $R_{bc} \in \mathbb{R}^{3 \times 3}$ is the rotation matrix in components in frame B that generates frame C .
- its 4th column p_{bc} is the homogeneous coordinates of the frame C origin expressed in frame B .
- it is the *absolute transformation* between the C and B coordinate frames (see paragraph just below figure 2.7 p. 45).

On the other hand, the interpretation of $g(\theta)$ is that it is the *relative transformation* between the C and B coordinate frames (generating C from B , using exponential coordinates).

The next cell calculates that the position vector of the origin of frame C relative to the origin of frame B , expressed in homogeneous coordinates, $p_{ab} - p_{ac}$ ("dum" in MATLAB code). Let us indicate the 1st three elements of p_{bc} as $p_{bc_{3 \times 1}}$. The next cell also shows that the 1st three elements of $p_{ab} - p_{ac}$ can also be obtained by transforming $p_{bc_{3 \times 1}}$ from being expressed in frame B to being expressed in frame A , namely $R_{ab} p_{bc_{3 \times 1}}$.

```
pab = gab(:,4); pac = gac(:,4); pbc = gbc(:,4);
```

```
dum = pac - pab
```

```
dum = 4×1
-1.8000
-1.0000
-0.1000
0
```

```
pb
```

```
pb = 4×1
-1.4473
-1.4678
0.0306
1.0000
```

```
shouldbezero3by1 = Rab*pb(1:3,1) - dum(1:3,1)
```

```
shouldbezero3by1 = 3×1
10-16 ×
0
0
-0.5551
```

The next cell shows that even though the norm of the 1st three elements of $p_{ab} - p_{ac}$ is the same as the norm of $p_{bc_{3 \times 1}}$, as expected, this is not the same as the norm of the 1st three elements of the last column of $g(\theta)$.

```
gbc
```

```
gbc = 4×4
-0.9117    0.2279    0.3419   -1.4473
-0.3221   -0.9131   -0.2501   -1.4678
0.2552   -0.3381    0.9059    0.0306
0         0         0         1.0000
```

```
gtheta
```

```
gtheta = 4×4
-0.9117    0.2279   -0.3419    1.7037
-0.3221   -0.9131    0.2501   -0.8154
-0.2552    0.3381    0.9059    0.4316
0         0         0         1.0000
```

```
norm(pb(1:3,1))
```

```
ans =
2.0616
```

```
norm(dum(1:3,1))
```

```
ans =
2.0616
```

```
norm(gtheta(1:3,4))
```

```
ans =
1.9375
```

What about g_{bc} ?

The twist coordinates of the transformation $g(\theta)$ was calculated above. The twist coordinates $\xi_b = (v_b, \omega_b)$ of g_{bc} can also be calculated, but these will not be in frame A , but in frame B , instead (as is also shown later).

$$g_{bc} = e^{\hat{\xi}_b \theta_b}$$

```
dum1 = rotm2axang(gbc(1:3,1:3))
```

```
dum1 = 1x4
      -0.1561    0.1538   -0.9757    2.8559
```

```
omega_b = dum1(1:3)'
```

```
omega_b = 3x1
      -0.1561
       0.1538
      -0.9757
```

```
omega
```

```
omega = 3x1
       0.1561
      -0.1538
      -0.9757
```

```
theta_b = dum1(4)
```

```
theta_b =
      2.8559
```

```
theta_b - theta
```

```
ans =
      4.4409e-16
```

We see that $\theta = \theta_b$. The next cell tests whether the calculated variables ω_b and θ_b are correct.

```
gbc(1:3,1:3)
```

```
ans = 3x3
      -0.9117    0.2279    0.3419
      -0.3221   -0.9131   -0.2501
       0.2552   -0.3381    0.9059
```

```
h_omega = hatm_num(omega_b);
shouldbezero4times4 = rodrigues(h_omega,theta) - gbc(1:3,1:3)
```

```
shouldbezero4times4 = 3x3
10-15 x
      -0.1110   -0.3053    0.1110
       0.3331   -0.4441         0
       0.2220    0.1110    0.1110
```

The next cell calculates v_b , using MLS eqn. (2.38) p. 43.

```
A = (eye(3) - gbc(1:3,1:3))*h_omega + omega_b*omega_b'*theta_b
```

```
A = 3x3
       0.3446    1.8500    0.6936
      -1.9736    0.3428   -0.0804
       0.0907   -0.6923    2.7322
```

```
v_b = A\gbc(1:3,4)
```

```
v_b = 3×1  
    0.6109  
   -0.8153  
   -0.2157
```

```
shoudlbezero3times = A*v_b - gbc(1:3,4)
```

```
shoudlbezero3times = 3×1  
10-16 ×  
     0  
     0  
    0.1735
```

The next cell calculates the twist as a 6×1 vector.

```
format long  
xi_b = [v_b;omega_b]
```

```
xi_b = 6×1  
    0.610891052923734  
   -0.815272410006449  
   -0.215657797351226  
   -0.156148523245089  
    0.153829166292768  
   -0.975681416388592
```

```
format default
```

The next cell test whether the twist $\xi_b = (v^b, \omega^b)$ or (v_b,omega_b) in MATLAB variables does actually produce g_{bc} .

```
dummat = gtwist_num(xi_b,theta_b)
```

```
dummat = 4×4  
   -0.9117    0.2279    0.3419   -1.4473  
   -0.3221   -0.9131   -0.2501   -1.4678  
    0.2552   -0.3381    0.9059    0.0306  
         0         0         0    1.0000
```

```
gbc
```

```
gbc = 4×4  
   -0.9117    0.2279    0.3419   -1.4473  
   -0.3221   -0.9131   -0.2501   -1.4678  
    0.2552   -0.3381    0.9059    0.0306  
         0         0         0    1.0000
```

```
mustbezero4times4 = dummat - gbc
```

```
mustbezero4times4 = 4×4  
10-15 ×  
   -0.1110   -0.7494    0.0555   -0.8882  
    0.7772   -0.4441   -0.0555    0.4441  
    0.2776    0.1665    0.1110    0.5343  
         0         0         0         0
```

According to the definition of the adjoint matrix, MLS p. 55, it is used to transform twists from the body frame B to the inertial frame A by

$$\xi = \text{Ad}_{g_{ab}} \xi^b,$$

where ξ and ξ^b are the twist coordinates in the inertial and the body frames, respectively.

```
xi_alt = adjmg(gab)*xi_b
```

```
xi_alt = 6x1
 0.5522
 0.8075
-0.0283
 0.1561
-0.1538
-0.9757
```

```
shouldbezero6times1 = xi - xi_alt
```

```
shouldbezero6times1 = 6x1
10-15 x
 0
-0.3331
 0.0729
 0.2220
-0.2220
 0.1110
```

This confirms that the twist ξ expressed in frame A is the same as the twist ξ_b expressed in frame B .

Screw associated with $g(\theta)$

Let's determine the screw associated with ξ and θ , where $g(\theta) = e^{\hat{\xi}\theta}$.

Since $\omega \neq 0$ and furthermore $\|\omega\| = 1$, according to MLS eq. (2.43) p. 47:

$$l = q + \lambda\omega,$$

where

$$q = \hat{\omega}v$$

is the coordinates of a point on the screw axis. The direction of the axis is given by ω .

The pitch of the screw is given by MLS eq. (2.42):

$$h = \omega^T v$$

and the magnitude by $M = \theta$ (see text just below MLS eq. (2.44)).

```
q = cross(omega,v)
```

```
q = 3x1
 0.7922
-0.5344
 0.2110
```

```
omega
```

```
omega = 3x1
 0.1561
-0.1538
-0.9757
```



```
h = omega'*v
```

```
h =  
-0.0104
```

```
Magnitude = theta
```

```
Magnitude =  
2.8559
```

Screw associated with g_{bc}

that is, the screw associated with $\xi_b = (v_b, \omega_b)$. Using the same calculation:

```
q_b = cross(omega_b,v_b) % Point on the screw axis
```

```
q_b = 3x1  
-0.8286  
-0.6297  
0.0333
```

```
omega_b
```

```
omega_b = 3x1  
-0.1561  
0.1538  
-0.9757
```

```
h_b = omega_b'*v_b
```

```
h_b =  
-0.0104
```

```
Magnitude = theta_b
```

```
Magnitude =  
2.8559
```

We see that the both the pitch and the magnitude of the screws associated with $g(\theta)$ and with g_{bc} are the same. Are the axes the same, though? The directions of the two twists are the same, because if ω_b (which is in frame B) is expressed in components in frame A , it is seen to be exactly the same as ω :

```
shouldbezero3by1 = omega - Rab*omega_b
```

```
shouldbezero3by1 = 3x1  
10-15 x  
0.2220  
-0.2220  
0.1110
```

To test whether the two axes are the same, we need to calculate the coordinates of point q_b on the axis of the screw associated with g_{bc} , which were calculated in frame B , in frame A , as "q_b_a" in the next code cell.

If a point q_2 lies on the axis of the screw associated with $g(\theta)$, $q_2 = q + \lambda\omega$, according to MLS equation (2.43) p. 47. Then,

$$\hat{\omega}q_2 = \hat{\omega}q + \lambda\hat{\omega}\omega = \hat{\omega}q.$$

For q_2 we shall here use q_b , but with its coordinates also in frame A .

```
q_b_a = gab*[q_b;1] % Coordinates of q_b in frame A
```

```
q_b_a = 4×1  
    0.8333  
   -0.5749  
   -0.0456  
    1.0000
```

```
shouldbezero3by1 = cross(omega,q) - cross(omega,q_b_a(1:3,1))
```

```
shouldbezero3by1 = 3×1  
10-15 ×  
   -0.2220  
         0  
   -0.0278
```

The previous code cell confirms that the two cross products are equal, which therefore means that point q_b lies on the axis of the screw associated with $g(\theta)$. Therefore all properties of the two screws are the same, which means that g_{bc} and $g(\theta)$ physically define the same screw.